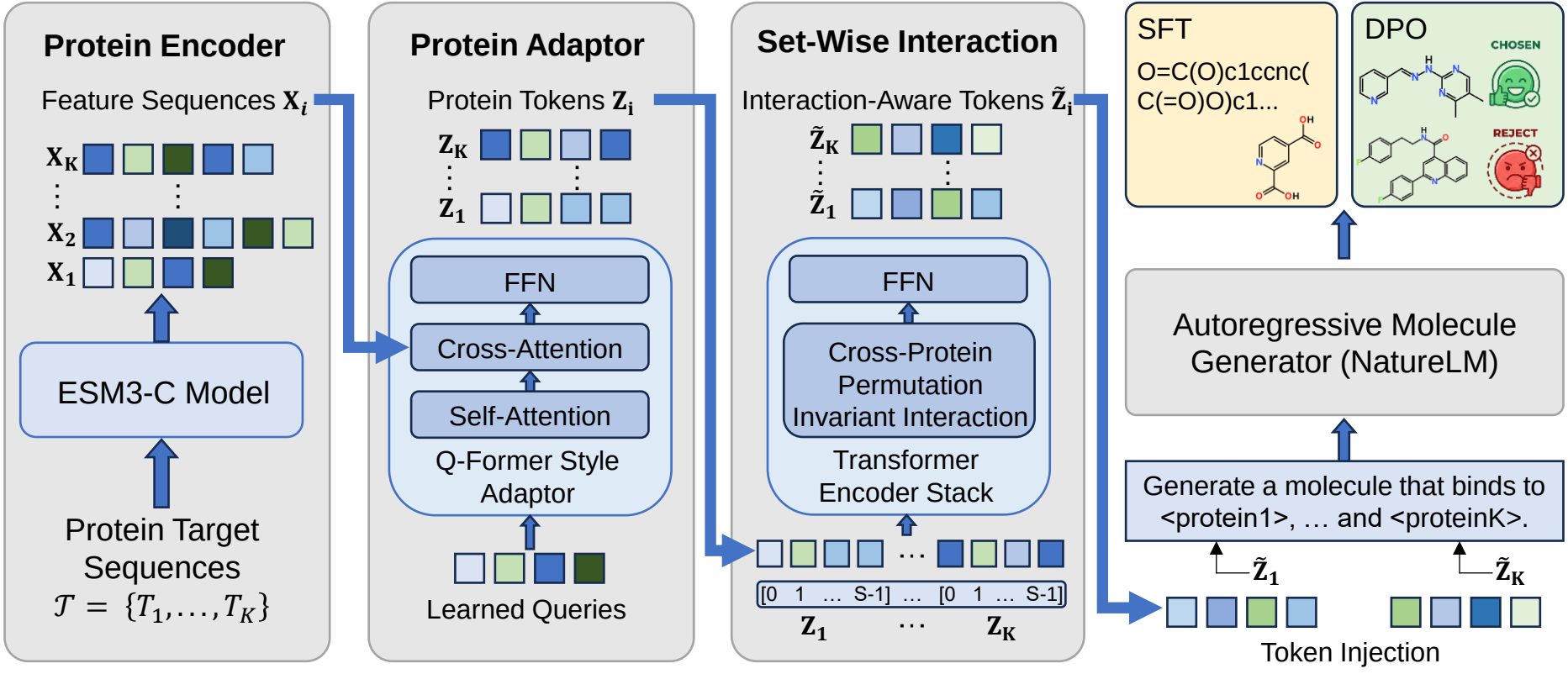

Protein Encoder
Feature Sequences $\mathbf{X}_i$
$\mathbf{X_K}$
$\mathbf{X_2}$
$\mathbf{X_1}$
ESM3-C Model
Protein Target Sequences
$\mathcal{T} = \{T_1, \ldots, T_K\}$
Protein Adaptor
Protein Tokens $\mathbf{Z}_i$
$\mathbf{Z_K}$
$\mathbf{Z_1}$
FFN
Cross-Attention
Self-Attention
Q-Former Style Adaptor
Learned Queries
Set-Wise Interaction
Interaction-Aware Tokens $\tilde{\mathbf{Z}}_i$
$\tilde{\mathbf{Z}}_K$
$\tilde{\mathbf{Z}}_1$
FFN
Cross-Protein Permutation Invariant Interaction
Transformer Encoder Stack
[0 1 … S-1] … [0 1 … S-1]
$\mathbf{Z_1}$ … $\mathbf{Z_K}$
SFT
O=C(O)c1ccnc(
C(=O)O)c1...
DPO
CHOSEN
REJECT
Autoregressive Molecule Generator (NatureLM)
Generate a molecule that binds to <protein1>, … and <proteinK>.
$\tilde{\mathbf{Z}}_1$
$\tilde{\mathbf{Z}}_K$
Token Injection